

# HR Attrition Analysis Project Documentation

TalentCore Solutions | Role: Junior Data Analyst | Domain: HR Analytics | Total Questions: 16

## BUSINESS PROBLEM

TalentCore Solutions is experiencing a rising employee attrition rate of approximately 16% annually. The HR department lacks visibility into who is leaving, why they are leaving, and when they tend to leave. Each lost employee costs the company over ₹50,000 in replacement costs. The board requires a data-driven retention strategy before the upcoming board meeting.

| WHO is leaving?         | WHY are they leaving?      | WHEN do they leave?         |
|-------------------------|----------------------------|-----------------------------|
| Age, Dept, Role, Gender | Salary, Balance, Promotion | Years, Career Stage, Tenure |

## BUSINESS QUESTIONS (16 Total)

| Question | Business Question                                                                                           | Category |
|----------|-------------------------------------------------------------------------------------------------------------|----------|
| Q1       | Which age group has the highest attrition?                                                                  | WHO      |
| Q2       | Which department has the highest attrition rate?                                                            | WHO      |
| Q3       | Which job role has the most attrition?                                                                      | WHO      |
| Q4       | Do employees with low monthly income leave more?                                                            | WHY      |
| Q5       | Which work life balance level has the highest attrition?                                                    | WHY      |
| Q6       | Does lack of promotion lead to attrition? (Uses YearsSinceLastPromotion)                                    | WHY      |
| Q7       | Do employees with no stock options leave more? (StockOptionLevel = 0)                                       | WHY      |
| Q8       | Which performance rating group leaves the most?                                                             | WHY      |
| Q9       | Which education field has the most attrition?                                                               | WHO      |
| Q10      | Does overtime lead to higher attrition? (Bonus question — strong signal)                                    | WHY      |
| Q11      | Does environment satisfaction affect attrition?                                                             | WHY      |
| Q12      | Do employees who worked at more companies leave sooner? (Uses NumCompaniesWorked)                           | WHEN     |
| Q13      | Which business travel frequency has the highest attrition rate? (Non-travel vs Travel Rarely vs Frequently) | WHO      |
| Q14      | After how many years at the company do employees leave the most? (Peak attrition year analysis)             | WHEN     |
| Q15      | Does having a different manager frequently lead to attrition? (Uses YearsWithCurrManager)                   | WHEN     |
| Q16      | What is the overall attrition rate and how many employees left? (Foundation metric for all analysis)        | BASE     |

## LEGEND

WHO - Identifies which group of employees are leaving

WHY - Identifies reasons behind attrition

WHEN - Identifies timing patterns of attrition

BASE - Foundation metric for the entire analysis

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